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import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

np.random.seed(42)

n = 500

study_hours = np.random.randint(1, 10, n)
attendance = np.random.randint(50, 100, n)

result = []

for i in range(n):
    if study_hours[i] >= 5 and attendance[i] >= 70:
        result.append(1)
    else:
        result.append(0)

df = pd.DataFrame({
    "Study_Hours": study_hours,
    "Attendance": attendance,
    "Pass": result
})

print(df.head())

X = df[["Study_Hours", "Attendance"]]
y = df["Pass"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
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test_size=0.2,
random_state=42
)
model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", round(accuracy * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
new_student = [[7, 85]]
prediction = model.predict(new_student)
print("\nStudy Hours:", new_student[0][0])
print("Attendance:", new_student[0][1])
if prediction[0] == 1:
    print("Prediction: PASS")
else:
    print("Prediction: FAIL")
```